

PROJECT NULLWEAR

Engineering Specification

Hardware design, mechanical drawings, power architecture, BoM and assembly specs

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NULLWEAR/P (Personal) · NULLWEAR/V (Vehicle) · NULLWEAR/S (Station)

Companion document to the Mitigation Report — full engineering detail for procurement and manufacture.

Document Revision A · Reference design · Subject to refinement during pilot phase.

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Classification

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*This document is the engineering companion to the strategic mitigation report on the Axon BLE OUI vulnerability.
All component values and dimensions are reference design and subject to refinement during pilot manufacture.
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Document Control

This document is the engineering specification for the NULLWEAR family of selective per-packet BLE annihilation devices, comprising the personal-issue NULLWEAR/P, the vehicle-mounted NULLWEAR/V, and the station-mounted NULLWEAR/S. It is a companion document to the strategic Mitigation Report and is intended for procurement officers, manufacturing partners, and engineering reviewers.

Field	Value
Document title	Project NULLWEAR — Engineering Specification
Document revision	Rev A — initial release
Status	Reference design; subject to refinement during pilot manufacture
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Classification	CONFIDENTIAL — Responsible Disclosure
Companion to	Mitigation Report — "Axon Bluetooth Vulnerability — A Mitigation for Law Enforcement"
Variants covered	NULLWEAR/P (Personal), NULLWEAR/V (Vehicle), NULLWEAR/S (Station)
Target manufacturer (initial)	Domestic Australian contract manufacturer; second-source US/EU
Target volume (Year 1)	75,000 units across all variants

PART I — PRODUCT OVERVIEW

What the device is, who carries it, and what it does to a BLE radio packet.

1. Product Definition

NULLWEAR is a passive-receive, reactive-transmit BLE radio device that performs **selective per-packet annihilation** of Bluetooth Low Energy advertising packets carrying a Media Access Control (MAC) address prefixed with the Axon Enterprise Organisationally Unique Identifier 00:25:DF. The device transmits no information of its own. It does not pair, advertise, or accept connections. Its sole purpose is to corrupt, in flight, the broadcasts of nearby Axon equipment, rendering them undetectable to any third-party BLE scanner in the vicinity.

The product is offered in three form factors, sharing a common silicon and firmware core:

- **NULLWEAR/P (Personal)** — body-worn, battery-powered, 50 × 35 × 11 mm, ~28 g, 18–24 h battery life.
- **NULLWEAR/V (Vehicle)** — vehicle-mounted, 12 V hard-wired, external antenna, 75 × 50 × 22 mm.
- **NULLWEAR/S (Station)** — wall- or DIN-rail-mounted, mains-powered, distributed antennas, 200 × 150 × 60 mm.

2. Operational Concept

The device is issued at three layers to provide complete physical coverage of an officer's operational envelope:

- **On the officer.** NULLWEAR/P is clipped to the duty belt or attached to the body-camera mount. It produces a 10–30 m radius protection bubble around the wearer, sufficient to cover any Axon equipment the officer is carrying.
- **In the vehicle.** NULLWEAR/V is wired into the patrol vehicle's 12 V switched power and connected to an external 2.4 GHz antenna, typically roof-rail-mounted. It provides a 30–50 m bubble around the vehicle, covering officers when they exit the vehicle and operate within the immediate vicinity.
- **In the station.** NULLWEAR/S mounts in the comms cabinet of every police station and drives a small number of distributed ceiling antennas (typically one per room of operational interest). The station as a whole is therefore inside a continuous protection zone, including briefing rooms, equipment lockers, charging docks, and muster areas where many Axon devices are present at once.

Together, the three layers ensure that an Axon device on a sworn officer is inside a protection bubble at every point in the duty cycle: in the locker, in the briefing, in the vehicle, on patrol, and back in the station.

3. Variants and Common Architecture

All three variants share the same radio core, the same firmware architecture, and the same Axon-OUI detection logic. The differences between them are in form factor, power source, antenna design, and protection-bubble radius. This shared architecture is essential: it means three production lines feed one inventory of silicon and one firmware build, with downstream variation only in mechanical and power subsystems.

AIR INTERFACE — 2.4 GHz BLE Advertising Channels (37, 38, 39)

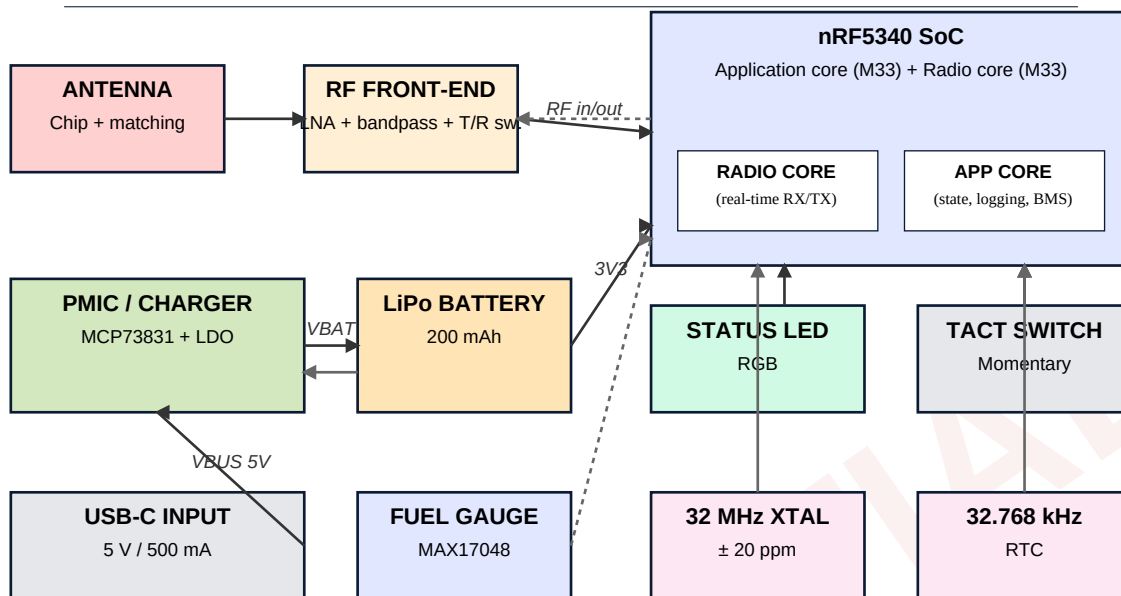


FIG 3 — NULLWEAR/P SYSTEM BLOCK DIAGRAM

Solid arrows = signal/power. Dashed = control.

Figure 3 — System block diagram common to all NULLWEAR variants. The radio core of the nRF5340 SoC handles the real-time RX/TX timing required for per-packet CRC corruption; the application core handles state, battery management, and logging.

PART II — FUNCTIONAL & PERFORMANCE REQUIREMENTS

What the device must do, measurably.

4. Detection Requirements (DR)

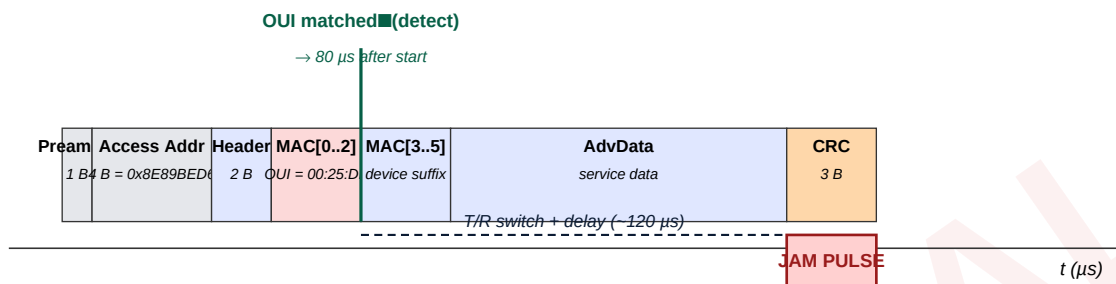
ID	Requirement	Acceptance criterion
DR-01	Continuously monitor BLE advertising channels 37, 38, 39	Channel hop interval $\leq 250\ \mu\text{s}$; 100% coverage of 3-channel advertising window over any 1 s
DR-02	Detect a packet with MAC prefix 0x00 0x25 0xDF within the first 80 μs of the packet	Verified by SDR-based test rig; $\geq 99.5\%$ true-positive rate at RSSI $\geq -85\ \text{dBm}$
DR-03	Distinguish 0x00 0x25 0xDF from any other OUI	False-positive rate $\leq 0.01\%$ across full BLE-band traffic in a populated lab environment
DR-04	Receive sensitivity at $\geq 1\ \text{Mbps}$ PHY	$\leq -96\ \text{dBm}$ at 0.1% PER (per nRF5340 datasheet)

5. Annihilation Requirements (AR)

ID	Requirement	Acceptance criterion
AR-01	Switch from RX to TX within 100 μs of OUI match	Verified at RF test rig; nRF5340 spec allows 40 μs with PPI/DPPI hardware path
AR-02	Emit collision pulse aligned with packet CRC region (last 24 bits)	Logic-analyser capture shows TX active during CRC bits with $\geq 80\%$ temporal overlap
AR-03	CRC-failure rate at receiver	$\geq 99\%$ of targeted packets discarded by reference receiver at distances up to 30 m
AR-04	No emission outside corruption window	Spectrum-analyser dwell shows ambient emission $\leq -90\ \text{dBm}$ in idle state
AR-05	Selective: no impact on non-OUI BLE traffic	Co-located Wi-Fi, Zigbee, and unrelated BLE devices show $< 1\%$ packet loss
AR-06	Maximum effective TX power	+8 dBm (within LIPD 2015 class licence and AS/NZS 4268 limits)

FIG 5 — BLE ADVERTISING PACKET CORRUPTION TIMING

Total advertisement ~210 μ s at 1 Mbps. Detection within first 80 μ s leaves 130 μ s for T/R + jam.



Result: CRC fails at receiver. Packet silently discarded.

Figure 5 — Timing budget for selective CRC corruption. A BLE 1 Mbps advertising packet is ~210 μ s in length. The device detects the OUI within 80 μ s of the preamble, switches from RX to TX within ~120 μ s, and emits a 24-bit duration collision pulse during the CRC trailer.

6. Power Requirements (PR)

The power budget for NULLWEAR/P is the hardest engineering constraint. It must run for an entire shift on a small battery without recharging, while keeping the radio in continuous receive on three channels. The nRF5340's hardware-managed scan/listen and DCDC efficiency make this achievable.

Variant	Source	Avg current	Peak	Endurance
NULLWEAR/P	200 mAh LiPo @ 3.7 V	~8 mA RX-continuous, ~10 mA mixed	17 mA peak (TX)	≥ 18 h cold-start; 24 h typical
NULLWEAR/V	12 V switched + 500 mAh LiPo backup	~25 mA from 12 V	60 mA peak	Indefinite while engine; 6 h on backup
NULLWEAR/S	100–240 V AC, internal 5 V SMPS, 2200 mAh UPS	~150 mA per antenna node	400 mA peak (cluster)	Indefinite mains; 4 h UPS

7. Environmental Requirements (ER)

ID	Parameter	IP	IV	IS
ER-01	Operating temperature	–10 to +55 °C	–20 to +70 °C	0 to +50 °C
ER-02	Storage temperature	–20 to +60 °C	–30 to +80 °C	–10 to +60 °C
ER-03	Ingress protection	IP67	IP65	IP30 (indoor)
ER-04	Drop test (concrete)	1.5 m, 6 faces	Vibration to ISO 16750-3	N/A
ER-05	Humidity	0–95% RH non-condensing	0–95% RH	0–80% RH

ID	Parameter	/P	/V	/S
ER-06	Salt-fog (coastal use)	MIL-STD-810G Method 509.6, 48 h	Same	Not required
ER-07	EMC emissions	AS/NZS CISPR 32 Class B	ISO 7637-2 (vehicle)	AS/NZS CISPR 32 Class B
ER-08	EMC immunity	AS/NZS 61000-4-3 @ 10 V/m	ISO 11452-2	AS/NZS 61000-4-3

PART III — RADIO SUBSYSTEM

RF chain, antenna design, and front-end specification.

8. Radio Architecture

The radio is built around the Nordic Semiconductor nRF5340, a dual-Cortex-M33 SoC with a dedicated 2.4 GHz radio peripheral on the network core. The radio core provides hardware support for sub-microsecond reactive transmission via the Programmable Peripheral Interconnect (PPI) and Distributed PPI (DPPI) — this is what makes the per-packet CRC corruption technique feasible at the timing required.

The nRF5340 was selected over the older nRF52840 (which is also capable) for three reasons: (a) the dedicated radio core means reactive timing is not pre-empted by application code; (b) lower active current draw at the same TX power, extending battery life by ~25%; (c) second-source from Nordic plus reference-design support from Nordic Semiconductor's Singapore office, simplifying domestic supply.

Stage	Component	Function	Spec
Antenna	Johanson 2450AT43A100 chip antenna	2.4 GHz radiation	Peak gain -1.5 dBi; bandwidth 2.4–2.5 GHz; PCB keepout 5 × 12 mm
Matching	0402 LC pi-network (tuned per board)	50 Ω match to RF feed line	Insertion loss ≤ 0.5 dB; return loss ≥ 10 dB across band
RF front-end	Skyworks SKY66112-11 FEM	LNA on RX path, PA bypass on TX path	LNA gain +10 dB, NF 1.5 dB; PA disabled (we use SoC TX directly)
T/R switch	Integrated in SKY66112-11	Hardware switch between RX and TX	Switching time ≤ 200 ns
SoC radio	nRF5340 internal 2.4 GHz transceiver	PHY layer; PPI-driven reactive TX	RX sensitivity -96 dBm @ 1 Mbps; TX +8 dBm

9. Antenna Design

NULLWEAR/P uses a chip antenna for size; NULLWEAR/V uses an external whip or magnetic-mount antenna for range; NULLWEAR/S uses a star topology of distributed indoor antennas connected via low-loss coax.

- **NULLWEAR/P:** Johanson 2450AT43A100 chip antenna on PCB top layer with mandatory 5 × 12 mm copper-keepout zone. Gain -1.5 dBi peak. Effective range 10–30 m line-of-sight.
- **NULLWEAR/V:** External whip antenna (Linx ANT-2.4-CW-RH or equivalent), magnetically mounted on roof rail or window glass via SMA pigtail. Gain +2 dBi. Effective range 30–50 m line-of-sight.
- **NULLWEAR/S:** Star topology: central unit with 4× to 8× SMA ports, each driving a ceiling-mounted dipole or patch antenna (Taoglas TG.30.8113 or equivalent) via RG-316 coax of length ≤ 10 m. Per-antenna effective range 5–15 m indoors.

PART IV — DIGITAL SUBSYSTEM & FIRMWARE

SoC, memory, firmware architecture, and the state machine.

10. SoC Selection

The Nordic nRF5340 is the reference SoC. Key properties:

- Dual ARM Cortex-M33: application core (128 MHz) + dedicated network core (64 MHz).
- 1 MB Flash + 512 kB RAM on application core; 256 kB Flash + 64 kB RAM on network core.
- 2.4 GHz radio peripheral on network core with PPI/DPPI hardware event chaining.
- TrustZone-M support for firmware-update authentication.
- QFN-94 package, 7 × 7 mm; commercial supply guaranteed through 2035.
- Second-source: nRF52840 (older, capable but with marginal timing) and Texas Instruments CC2652R7 (alternative architecture).

11. Memory Layout

```
APP CORE (1 MB Flash, 512 kB RAM)
0x00000000 - 0x0000C000 MCUboot bootloader (signed)
0x0000C000 - 0x000F0000 Application image slot 0 (active)
0x000F0000 - 0x000F8000 Application image slot 1 (OTA staging) – N/A on /P
0x000F8000 - 0x00100000 Settings + non-volatile event log

NETWORK CORE (256 kB Flash, 64 kB RAM)
0x01000000 - 0x01040000 Radio firmware (PPI handlers, OUI matcher, jam pulse)

RAM USAGE
Stack ..... 32 kB
Heap ..... 16 kB
Logging ring . 16 kB
Free ..... 448 kB
```

12. Firmware Architecture

Firmware is partitioned across the two cores. The radio core runs a minimal real-time loop that owns the RF state machine and the per-packet detection / corruption pipeline. The application core handles battery management, LED state, fault detection, and any optional logging. The two cores communicate via Nordic's shared-memory IPC.

FIG 9 — NULLWEAR FIRMWARE STATE MACHINE

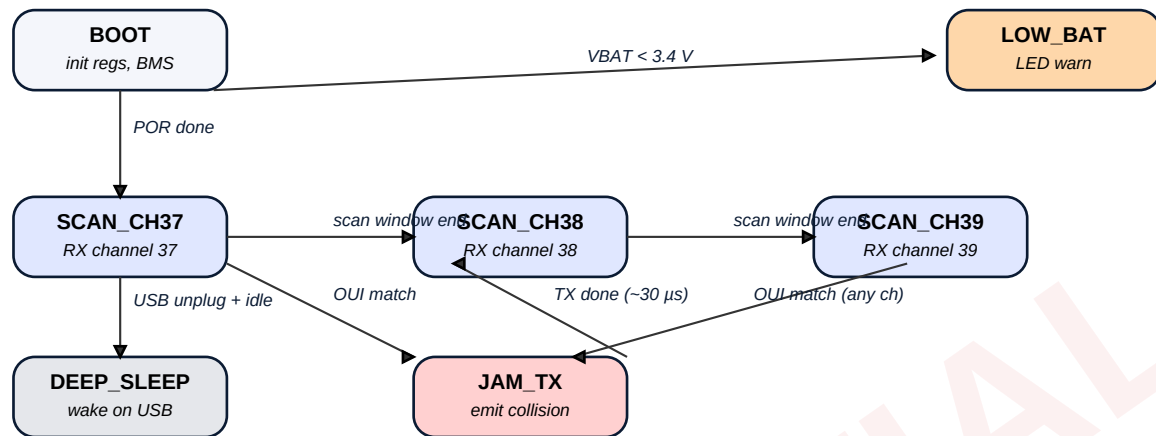


Figure 9 — Top-level firmware state machine. The three SCAN states (one per advertising channel) cycle every ~80 ms; OUI match in any channel triggers the JAM_TX state, which returns to scanning after the ~30 μs corruption pulse.

12.1 Detection ISR (radio core)

The radio core's PPI/DPPI subsystem is configured to fire an interrupt after the radio peripheral has received the packet's access address (32 bits) and the first 24 bits of the MAC. The ISR compares the 24 bits to 0x00 0x25 0xDF in a single CMP instruction. If matched, the ISR triggers a hardware-chained TX preparation sequence:

```

// Radio core ISR (pseudo-C, runs on packet header complete)
void RADIO_IRQHandler(void) {
  uint32_t mac_first24 = NRF_RADIO->RXMATCH; // first 24 bits of MAC
  if ((mac_first24 & 0xFFFFF) == 0x002500DF) { // little-endian compare
    NRF_RADIO->TASKS_DISABLE = 1;
    NRF_RADIO->PACKETPTR = (uint32_t)&jam_pdu;
    NRF_RADIO->TASKS_TXEN = 1;
    // PPI chains TXREADY -> START -> END -> RXEN automatically
  }
  NRF_RADIO->EVENTS_RXMATCH = 0;
}

```

12.2 Battery management (app core)

- Polls the MAX17048 fuel gauge every 60 s via I²C.
- Drives the RGB status LED: green (>50%), amber (20–50%), red (<20%), red-blink (<5%).
- Logs charge cycles, deep-discharge events, and abnormal temperatures to NVM.
- Initiates ordered shutdown if VBAT < 3.0 V; locks out wake-from-USB if cell temperature exceeds 60 °C.

PART V — POWER SUBSYSTEM

Battery chemistry, charging, power management, and life calculations.

13. Power Architecture

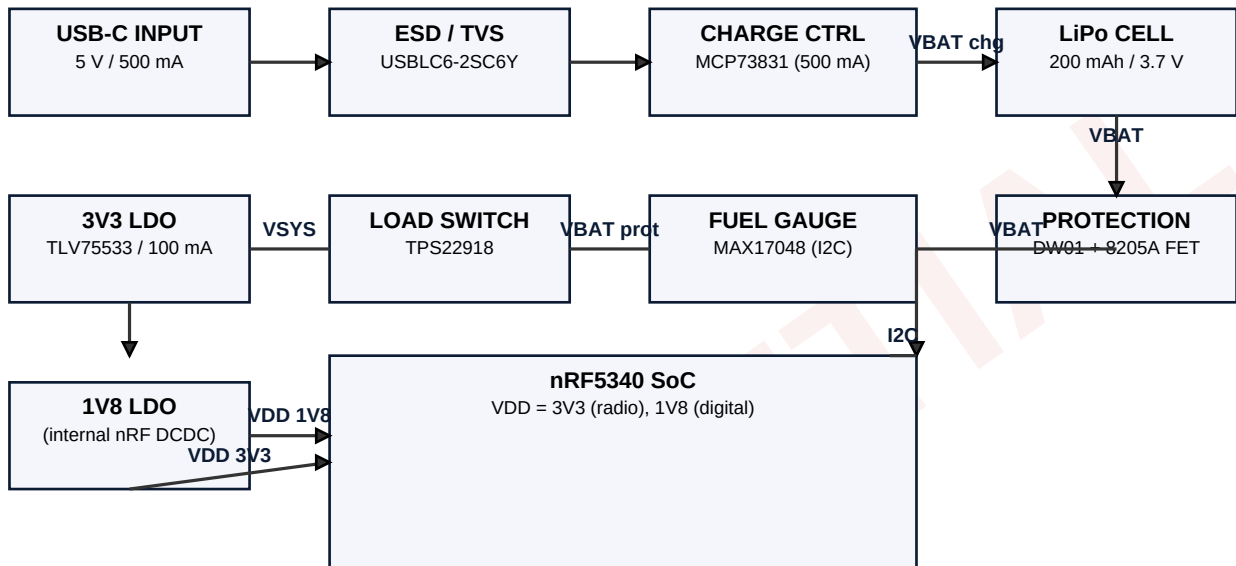


FIG 4 — NULLWEAR/P POWER ARCHITECTURE

Figure 4 — Power architecture for NULLWEAR/P. USB-C input feeds an MCP73831 single-cell LiPo charger via a TVS diode. The protected battery output drives a TPS22918 load switch and a TLV75533 LDO supplying 3V3 to the SoC. The nRF5340's internal DCDC generates 1V8.

14. Battery Chemistry & Selection

The reference cell for NULLWEAR/P is a 200 mAh single-cell lithium polymer (LiPo), 3.7 V nominal, with integrated PCM (protection circuit module). Cell selection criteria:

- **Chemistry:** Li-CoO $\blacksquare$ LiPo. Standard for wearables; high energy density (~150 Wh/kg).
- **Capacity:** 200 mAh nominal; 180 mAh at end-of-life (after 500 charge cycles to 80% retention).
- **Form factor:** 30 × 25 × 4 mm prismatic. Fits under PCB.
- **Discharge rate:** 1C continuous, 2C peak — well within the 17 mA peak draw.
- **Operating temperature:** charge 0–45 °C, discharge –10 to +60 °C.
- **PCM features:** overcharge cut-off at 4.25 V, over-discharge cut-off at 2.5 V, short-circuit cut-off at 4 A, integrated thermistor for temperature monitoring.
- **Suppliers:** PowerStream PGEB0042530 (US), Shenzhen Lipower LP402530 (CN), or domestic re-pack via Battery World industrial division (AU).

15. Charging Subsystem

USB-C input feeds a Microchip MCP73831 single-cell LiPo charger configured for 500 mA constant-current / 4.2 V constant-voltage. ESD protection is provided by a STMicroelectronics USBLC6-2SC6Y TVS diode.

Reverse-polarity protection on the USB-C input is provided by a P-MOSFET (Diodes DMP3098L). The charger asserts STAT to the SoC for charge-state indication; a fuel gauge (Maxim MAX17048) provides accurate state-of-charge over I²C.

Charge time from empty to 100% at standard ambient: 50 minutes. Charge time to 80% (sufficient for a full shift): 30 minutes.

16. Power Budget & Battery-Life Calculations

The continuous-receive load is dominated by the radio peripheral's RX current plus a small DCDC-conversion overhead. Average current draw of the entire system is calculated below:

Subsystem / state	Current	Duty cycle	Avg contribution
Radio RX (1 Mbps PHY, channel hop)	4.6 mA	98%	4.5 mA
Radio TX peak during corruption	16 mA	0.5%	0.08 mA
Radio T/R switching transient	12 mA	1.5%	0.18 mA
App-core CPU (event handling)	3 mA	5%	0.15 mA
App-core deep sleep	3 μ A	95%	~0 mA
LDO + DCDC quiescent	0.4 mA	100%	0.4 mA
Fuel gauge MAX17048	23 μ A	100%	0.02 mA
LED (green idle, dim)	0.5 mA	10%	0.05 mA
Total average current			5.4 mA
Including 50% RF environment contention			~8 mA typical

Battery life calculation:

Useful capacity = 200 mAh $\times$ 0.85 (PCM cut-off margin) = 170 mAh
 Average current = 8 mA (typical mixed-traffic environment)
 Endurance = 170 / 8 = 21.25 hours per charge
 Worst-case (high-traffic, peak 12 mA avg) = 170 / 12 = 14.2 hours
 Best-case (low-traffic, 6 mA avg) = 170 / 6 = 28.3 hours
 Specification: $\geq$ 18 hours per charge across all duty environments.

17. NULLWEAR/V Power Subsystem

The vehicle variant is hard-wired to the patrol vehicle's switched 12 V supply via a 1 A fast-blow fuse and a TVS diode (SMAJ18A) for load-dump protection per ISO 7637-2. A wide-input buck converter (TI LMR33630) steps 12 V down to 5 V at 90% efficiency. The 5 V rail feeds the same MCP73831/200 mAh LiPo backup-charger circuit as the personal variant; the LiPo provides ride-through for short engine-off intervals ($\leq$ 6 hours). The remainder of the power tree is identical to NULLWEAR/P.

Vehicle reverse-polarity protection: P-MOSFET (Infineon BSP254). Surge protection: MOV across 12 V input. Quiescent draw with engine off and battery full: $\leq$ 200 μ A, ensuring no measurable drain on the vehicle battery.

18. NULLWEAR/S Power Subsystem

The station variant is mains-powered via an internal universal-input AC-DC switching power supply (Mean Well IRM-30-5, 30 W, 5 V output). A 2200 mAh LiFePO₄ UPS battery (chosen for long calendar life and high cycle count) provides ~4 hours of continuous operation during mains outages. The unit drives multiple antenna nodes via active RF distribution, each node drawing ~150 mA at 5 V.

PART VI — MECHANICAL DESIGN

Enclosure, materials, mounting, and ingress protection.

19. Enclosure Materials

Variant	Material	Process	Finish
NULLWEAR/P	Polycarbonate / ABS blend (PC-ABS, UL94 V-0)	Injection-moulded, ultrasonic-welded seam, potted internally	Matte black, soft-touch coating
NULLWEAR/V	Polycarbonate (PC, UL94 V-0) with EPDM gasket	Injection-moulded, M3 stainless screws on gasketed flange	Matte black, UV-stabilised
NULLWEAR/S	Aluminium 6061-T6 extrusion with steel end caps	Extruded body, machined end caps, anodised	Black anodised, DIN-rail clip on rear

20. Form Factor — NULLWEAR/P

FIG 1 — NULLWEAR/P TOP VIEW (PCB-side, scale 4:1)

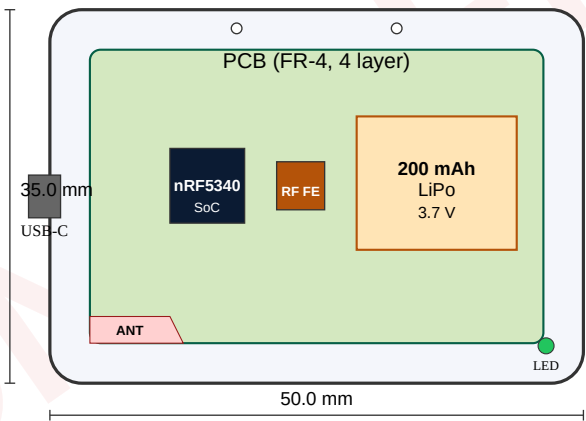


Figure 1 — NULLWEAR/P PCB-side top view. PCB shown in green; SoC, RF front-end, battery, USB-C, and antenna keepout indicated. Belt-clip mount holes shown along bottom edge.

FIG 2 — NULLWEAR/P SIDE VIEW (cross-section, scale 5:1)

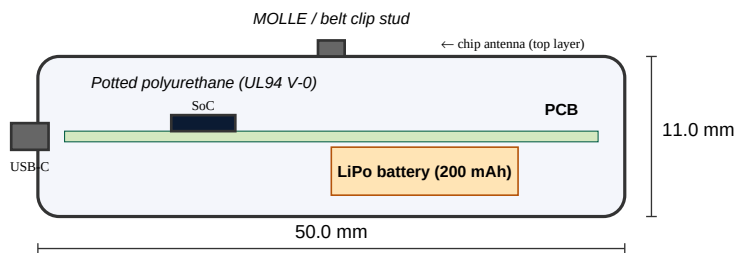


Figure 2 — NULLWEAR/P side cross-section. Battery sits below the PCB; SoC and RF components on PCB top side. Internal void is potted with polyurethane after assembly for shock and ingress protection.

Parameter	Specification
External dimensions	50 mm (L) × 35 mm (W) × 11 mm (H)
Mass	28 g ± 2 g
PCB outline	46 × 31 mm, 4-layer FR-4, 1.2 mm thickness
Antenna	Chip antenna on PCB top layer with 5 × 12 mm keepout
Battery	200 mAh LiPo, 30 × 25 × 4 mm, fitted under PCB
USB-C port	Right-angle SMD receptacle, sealed via internal gasket
LED	0805 RGB SMD, light-piped to enclosure top via 3 mm acrylic rod
Switch	Single momentary tact switch, recessed 0.5 mm under enclosure surface (button cover)
Mounting	Two M3 brass inserts on rear face for belt clip / MOLLE mount; lanyard pass-through hole
Ingress protection	IP67 (1 m water immersion for 30 min) achieved by ultrasonic weld + USB-C gasket + potting
Drop resistance	1.5 m drop onto concrete on each of 6 faces; no functional or cosmetic damage

21. Form Factor — NULLWEAR/V

Parameter	Specification
External dimensions	75 mm (L) × 50 mm (W) × 22 mm (H), excluding mounting flange
Mass	110 g (without antenna lead)
Mounting	4× M4 stainless screws through gasketed mounting flanges
Power input	Bare-wire pigtail with crimp ring terminals — fused 1 A in-line, with red (12 V), black (GND), white (ignition-sense) leads, 2 m length
Antenna connector	Female SMA jack with weatherproof rubber cap
Antenna (supplied)	Linx ANT-2.4-CW-RH whip with magnetic base; 3 m RG-316 coax with male SMA
Status indicators	Two LEDs visible through enclosure top: green (power), blue (RF active)
Ingress protection	IP65 (jet-spray) — suitable for engine-bay or under-dash mounting
Vibration / shock	ISO 16750-3 rigid-mount profile; 100 G shock per axis

FIG 7 — NULLWEAR/V VEHICLE INSTALLATION

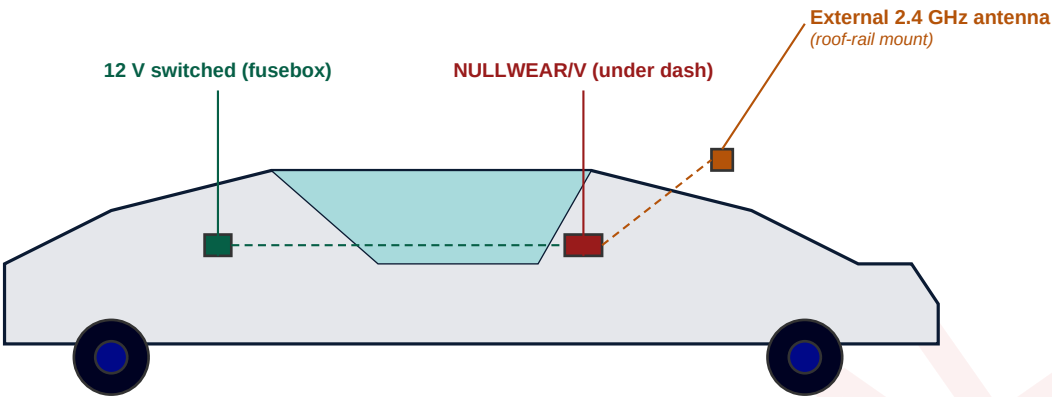


Figure 7 — Typical vehicle installation. NULLWEAR/V mounts under the dash on the passenger side; 12 V switched power is taken from a spare fusebox circuit; the external antenna mounts on the vehicle's roof rail or mirror via a magnetic base.

22. Form Factor — NULLWEAR/S

Parameter	Specification
External dimensions	200 mm (L) × 150 mm (W) × 60 mm (H)
Mass	1.6 kg
Mounting	DIN rail (35 mm) clip on rear, plus four M5 wall-mount holes
Power input	IEC C14 inlet, 100–240 V AC, 50/60 Hz, 30 W
Antenna ports	8× female N-type connectors on front panel
Cooling	Convection-only via finned aluminium top; no fan
Indicators	Front-panel display: power, mains/UPS, per-port RF activity (1× green LED per port)
Network	RJ-45 Ethernet for remote status only (no command surface)
UPS battery	Internal 2200 mAh LiFePO ₄ pack, hot-swappable via panel cover

FIG 8 — NULLWEAR/S DISTRIBUTED-ANTENNA INSTALLATION

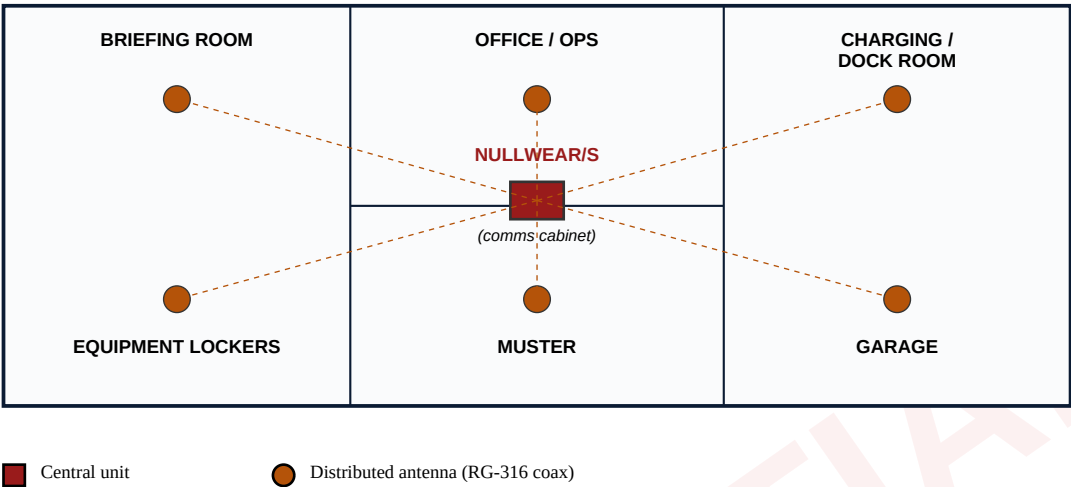


Figure 8 — Distributed-antenna installation in a typical police station. The central NULLWEAR/S unit lives in the comms cabinet; up to 8 ceiling-mounted antennas, fed by RG-316 coax, provide continuous coverage of every operational room in the building.

FIG 6 — FORM FACTOR COMPARISON (1:2 scale)

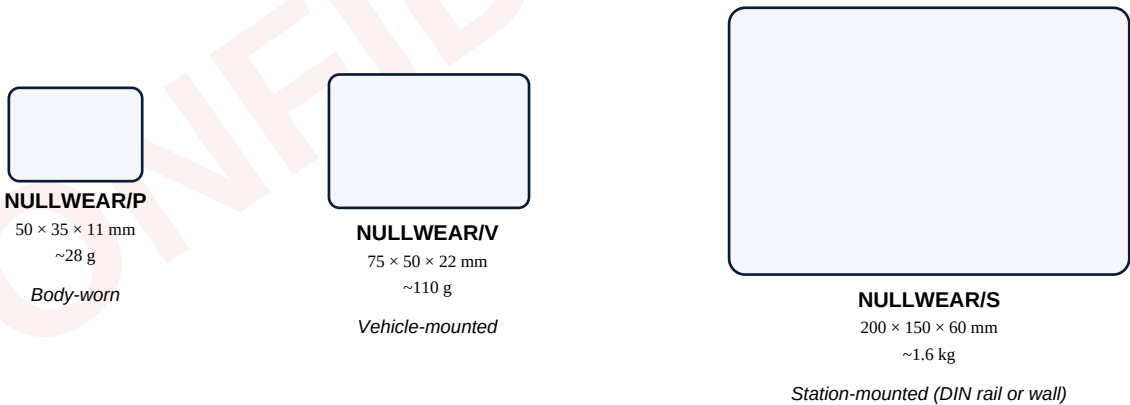


Figure 6 — All three NULLWEAR variants shown to scale (1:2). The personal device is the size of a credit-card stack; the vehicle variant is the size of a deck of playing cards; the station unit is the size of a small router.

PART VII — PCB & ASSEMBLY

PCB stack-up, layout constraints, and SMT manufacturing.

23. PCB Stack-up

FIG 10 — PCB STACK-UP (4-LAYER, 1.2 mm OVERALL)

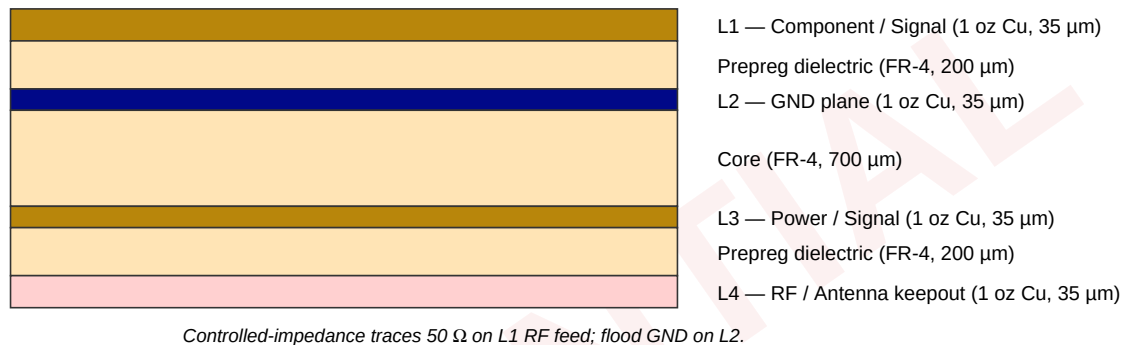


Figure 10 — PCB stack-up for NULLWEAR/P. 4-layer construction with continuous GND plane on L2 for RF return-current management; controlled-impedance 50 Ω trace from SoC RF pad to chip antenna.

24. Layout Constraints

- Chip antenna keepout area must contain no copper, vias, or components within 5 × 12 mm above the antenna footprint on all four layers.
- RF trace from SoC to RF front-end and from front-end to chip antenna is 50 Ω controlled impedance, microstrip with reference to L2 GND plane, length matched ≤ 0.5 mm.
- 32 MHz crystal must be within 3 mm of the SoC XC1/XC2 pins, with a guarded ground ring.
- Power-supply decoupling: 100 nF + 4.7 μF X7R MLCC pair within 1.5 mm of every SoC supply pin.
- Charger thermal pad connected to a 100 mm² internal copper pour for heat spreading.
- USB-C ESD diode placed within 2 mm of the USB-C receptacle, with low-inductance return path to GND.

25. SMT Assembly Process

- Solder paste: SAC305 lead-free, Type 4 mesh, applied via stainless steel laser-cut stencil (125 μm thickness).
- SoC (QFN-94) placement: high-precision pick-and-place required; bottom-side termination requires X-ray inspection for void analysis ($\leq 25\%$ void area on thermal pad).
- Reflow profile: ramp-soak-spike per JEDEC J-STD-020, peak 245 $^{\circ}\text{C} \pm 5$ $^{\circ}\text{C}$, time above liquidus ≤ 90 s.
- Through-hole (USB-C, RF connector, switch, LED light pipe): selective wave or hand-solder after SMT reflow.
- Conformal coating: parylene-C, 5–10 μm thick, applied via vapour deposition before potting.
- Potting: Wevolt PU-7000 polyurethane, two-part, fills internal void to 95% volume to displace air and provide shock isolation. UL94 V-0 rated.

PART VIII — BILL OF MATERIALS

Every component, every supplier, every cost line.

26. NULLWEAR/P — Bill of Materials (per unit)

Costs are reference prices in AUD at 50,000-unit production volume from a domestic Australian contract manufacturer. Component sources are dual-sourced where commercially available.

Ref	Description	MPN	Qty	Unit (AUD)	Ext (AUD)
U1	Dual-core M33 SoC + 2.4 GHz radio	Nordic nRF5340	1	\$8.40	\$8.40
U2	RF front-end module (LNA + T/R)	Skyworks SKY66112-11	1	\$2.10	\$2.10
U3	LiPo charger (CC/CV, 500 mA)	Microchip MCP73831T-2ACI	1	\$0.70	\$0.70
U4	LiPo fuel gauge (I ² C)	Maxim MAX17048G+T10	1	\$1.30	\$1.30
U5	3V3 LDO, 100 mA, low Iq	TI TLV75533PDBVR	1	\$0.45	\$0.45
U6	Load switch, low Rdson	TI TPS22918DBVR	1	\$0.55	\$0.55
D1	ESD/TVS for USB-C, 5-line	ST USBLC6-2SC6Y	1	\$0.40	\$0.40
Q1	Reverse-polarity P-MOSFET	Diodes DMP3098L-7	1	\$0.25	\$0.25
Y1	32 MHz crystal, ±20 ppm	Abracon ABM3B-32.000MHZ	1	\$0.55	\$0.55
Y2	32.768 kHz crystal (RTC)	Abracon ABS07-32.768KHZ	1	\$0.40	\$0.40
ANT1	2.4 GHz chip antenna	Johanson 2450AT43A100	1	\$0.95	\$0.95
BAT1	LiPo cell, 200 mAh, 30×25×4 mm with PCM	PowerStream PGEB0042530 / 2nd-source LP402530	1	\$3.80	\$3.80
J1	USB-C right-angle receptacle, sealed	Amphenol 12401610E4#2A	1	\$0.85	\$0.85
LED1	RGB LED, 0805	Cree CLV1A-FKB	1	\$0.30	\$0.30
SW1	Tactile switch, sealed	C&K; KMR2	1	\$0.35	\$0.35
	Passives (MLCC, resistors, inductors): est. 35 components	Multiple	35	\$0.04	\$1.40
	PCB: 4-layer FR-4, 1.2 mm, ENIG finish	Domestic / 2nd-source CN	1	\$2.40	\$2.40
	Enclosure (PC-ABS injection-moulded, 2-piece)	Custom tooled, AU mfr	1	\$3.80	\$3.80
	Belt clip (stainless, 304)	Custom	1	\$1.20	\$1.20
	Acrylic light pipe (3 mm rod, 12 mm)	Custom	1	\$0.30	\$0.30
	Polyurethane potting compound (per fill)	Wevolt PU-7000	1	\$1.10	\$1.10
	EPDM gasket strip, USB-C seal	Generic	1	\$0.20	\$0.20
	M3 brass insert (heat-staked) × 2	Generic	2	\$0.10	\$0.20
	Subtotal — components & enclosure				\$31.95
	Assembly labour (SMT + final + test)				\$8.50

Ref	Description	MPN	Qty	Unit (AUD)	Ext (AUD)
	Test, calibration, programming				\$3.00
	Packaging (individual blister + outer carton)				\$1.20
	Margin, overhead, contingency (~25%)				\$11.20
	Total NULLWEAR/P unit cost (50k volume)				\$55.85

27. NULLWEAR/V — Bill of Materials (delta from /P)

NULLWEAR/V uses the same SoC, RF chain, and firmware as NULLWEAR/P. The bill-of-materials delta below lists only items that differ from /P:

Ref	Description	MPN	Unit (AUD)
U10	Wide-input buck converter, 12 V → 5 V	TI LMR33630ADDAR	\$2.40
D2	Load-dump TVS diode (ISO 7637-2)	Littelfuse SMAJ18A	\$0.45
F1	Inline blade fuse, 1 A fast-blow + holder	Littelfuse FKS	\$0.80
BAT2	LiPo backup cell, 500 mAh	PowerStream LP503450	\$5.40
ANT-V	External whip antenna, magnetic base, 3 m coax	Linx ANT-2.4-CW-RH	\$11.00
J2	Female SMA panel-mount with cap	Amphenol 132270	\$3.20
	Heavier enclosure (gasketed PC, IP65 flange)	Custom	\$8.50
	Power harness pre-assembled (2 m, fused)	Custom	\$4.20
	NULLWEAR/V additional cost over /P		\$35.95
	Total NULLWEAR/V unit cost (25k volume)		\$82–95

28. NULLWEAR/S — Bill of Materials (delta)

Description	Notes	Unit (AUD)
Aluminium 6061-T6 enclosure (extruded body, machined ends, anodised)	Tooling amortised; per-unit cost at 1k volume	\$48.00
Mean Well IRM-30-5 universal-input AC-DC SMPS	30 W, 5 V, IEC C14	\$22.00
LiFePO ₄ UPS battery pack, 2200 mAh, 4 cells	With BMS and hot-swap connector	\$45.00
8× SoC sub-boards (one per antenna node)	Each ≈ /P less battery	\$210.00
8× N-type female panel connectors	With caps	\$24.00
8× indoor antennas (Taoglas TG.30.8113)	Or equivalent	\$56.00
8× RG-316 coax cables, 8 m each, with N-male	Pre-terminated	\$72.00
Front-panel display + RJ-45 monitoring board	Custom	\$28.00
Final assembly, test, calibration, packaging, shipping		\$95.00
Total NULLWEAR/S unit cost (1k volume)		\$600.00

PART IX — MANUFACTURING & QUALITY ASSURANCE

Production process and acceptance test procedures.

29. Production Workflow

- **Stage 1 — PCB fabrication.** 4-layer FR-4 boards manufactured to IPC-6012 Class 2. ENIG surface finish for solderability. AOI (automated optical inspection) at 100%.
- **Stage 2 — SMT placement.** All surface-mount components placed via high-precision pick-and-place with vision alignment. Reflow per JEDEC J-STD-020. AOI + AXI (X-ray inspection of QFN thermal pad voids) post-reflow.
- **Stage 3 — Through-hole and final solder.** USB-C, RF connector, switch, LED light pipe soldered via selective wave or hand. Visual inspection.
- **Stage 4 — Initial functional test.** Bare PCB tested on bed-of-nails fixture: power-on, supply rails, SoC programming, JTAG continuity.
- **Stage 5 — Battery installation.** LiPo connected via JST PH-2 connector with hot-glue strain relief. PCB seated into lower enclosure half.
- **Stage 6 — RF acceptance test.** Each unit tested in shielded RF chamber against a reference Axon BLE source: must achieve $\geq 99\%$ packet-discard rate at 5 m distance over 100 packet samples.
- **Stage 7 — Conformal coating + potting.** Parylene-C deposited via vapour. Polyurethane potting injected to fill internal void. Cured 4 hours at 40 °C.
- **Stage 8 — Enclosure closure.** Upper enclosure half ultrasonically welded to lower; USB-C gasket seated; belt clip / lanyard hardware fitted.
- **Stage 9 — Ingress test.** Sample-rate IP67 immersion test (1 m for 30 min) at 1% AQL.
- **Stage 10 — Final firmware load + serial provisioning.** Production firmware flashed via USB-C; unique serial number written to NVM and laser-etched on enclosure.
- **Stage 11 — Battery conditioning.** Initial charge cycle to $\sim 70\%$ state-of-charge for shipping (per IATA dangerous-goods regulations on lithium battery transport).
- **Stage 12 — Packaging.** Individual unit in moulded blister with quick-start card; outer carton holds 25 units.

30. Acceptance Test Procedure (per unit)

Test	Pass criterion
Power-on self-test	All rails within $\pm 5\%$; SoC boots within 1.2 s; LED green for ≥ 2 s
Battery state	VBAT > 3.7 V; reported SOC > 65%; no temperature warning
RX sensitivity	Detect Axon-OUI test packet at -90 dBm with $\geq 99\%$ capture rate
CRC corruption	Reference receiver records $\geq 99\%$ packet discard at 5 m over 100 packets
False-positive isolation	Co-located non-Axon BLE devices show < 1% packet loss
Quiescent emission	Spectrum-analyser dwell shows ambient TX ≤ -90 dBm in idle state
Charge cycle	Charges from 50% to 100% in ≤ 35 min; no thermal warning
Sleep current	Idle (no traffic) draw ≤ 100 μ A
Mechanical	Visual inspection: no enclosure damage, weld seam intact, light pipe flush

31. Quality Assurance

- ISO 9001:2015 quality-management system at the contract manufacturer.
- Sample-rate environmental testing per Part II — Environmental Requirements (1% AQL on IP67, 1% AQL on drop test, 0.1% AQL on temperature cycling per batch).
- Component traceability: every batch of SoCs and batteries is logged against unit serial numbers in a manufacturer-side database, retained for 10 years.
- Field-failure analysis: defective units returned to a Tier-1 reverse-logistics depot for fault analysis; root-cause findings feed back into the manufacturing process within 30 days.

PART X — OPERATIONS

Issue, daily use, charging, fault handling, and end-of-life.

32. Issue Procedure

- Each NULLWEAR/P is issued one-per-officer at the next regular equipment-issue checkpoint.
- Unit is registered to the officer's badge number in the agency asset-management system.
- Officer receives a one-page laminated quick-start card (in equipment locker).
- First-time training: 5-minute briefing during the next shift handover. No formal certification required.
- Unit is stored on the officer's existing body-camera dock during off-shift hours; the dock provides power to the USB-C input.

33. Daily Operation

- On clip-on, officer presses the front-panel button briefly. LED illuminates green to indicate normal operation.
- Unit operates entirely autonomously throughout the shift. No officer interaction is required.
- LED indications: **green steady** = battery > 50%, normal operation; **amber steady** = battery 20–50%; **red blink** = battery < 20%, charge before next shift; **blue flicker** = active corruption event in progress (occasional, not continuous).
- On removal, officer re-docks the unit. No power-down required; the unit auto-detects USB charge and enters charge mode.
- Total daily officer interaction time: under 30 seconds.

34. Fault Handling

- **Battery fault** (red blink during operation): officer reports at end of shift; replacement unit issued; faulty unit returned to depot.
- **No LED on power-on**: officer reports immediately; replacement unit issued; faulty unit returned.
- **USB-C charge fault** (no charge after 1 hour on dock): same as above.
- **No fault indication possible at officer level beyond LED state**. All deeper diagnostics performed at depot level via USB-C developer interface.

35. Maintenance and Lifecycle

- **Battery service life**: 500 charge cycles to 80% capacity retention. Typical agency use of 1 charge per shift = approximately 2 years to 80% retention.
- **Battery replacement**: not field-serviceable. Expired units returned to depot for refurbishment; new battery pack installed; potting partially restored; unit re-tested to acceptance procedure.
- **Firmware update**: not anticipated during normal service life; if required, performed at depot via USB-C bootloader. No OTA capability — by design, to eliminate remote attack surface.
- **Total expected service life**: 7+ years with one mid-life battery refurbishment.
- **End-of-life**: lithium battery extracted for recycling per AS/NZS 5377; PCB assemblies sent to e-waste recycling stream.

End of Engineering Specification.

Prepared by **Benjamin Jack Leighton** on behalf of **Tester Present Specialist Automotive Solutions**. Issued 05 May 2026. CONFIDENTIAL — Responsible Disclosure. Document Revision A.

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